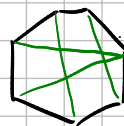


inlec: (A) first prove that zeros imply a diagram must have an ear

(B) then remove ear, discard the constraints, prove again that we must have an ear

- (A):
- (1) show that any surviving diagram must have a short
 - (2) show that any surviving diagram must have crossings with a short



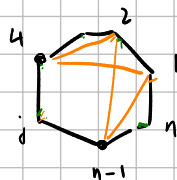
(A1) inlec: using the property that every edge must be part of a triangle:

look at edge $(i, i+1)$: must have either $X_{i, i+2}$ or $X_{i-1, i+1}$ or both $X_{i, i}$, $X_{i+1, i}$

$X_{i, i+1}$, $X_{i-1, i+1}$, $X_{i+1, i+3}$, $X_{i+n-2, i}$ are all shorts
 $(i \in \{i+3, \dots, i+n-2\})$

$\Rightarrow$ if we disallow shorts, for each edge $(i, i+1)$ we must have both chords $X_{i, i}$, $X_{i+1, i}$ for some $i \in \{i+4, \dots, i+n-3\}$

$\Rightarrow$ for each edge we must put it in a triangle that doesn't have shorts.



inlec: we cannot do this with $n-3$ chords.

$\Rightarrow$ so far: tried with Mathematica $\Rightarrow$ minimum number of chords is n

$\Rightarrow$

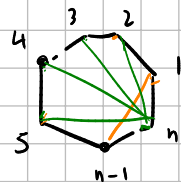
(2) NLOG assume short is $X_{i, n-1}$. Things that cross it are $X_{i, n}$, $i \in 2, \dots, n-2$

Rewrite w.r.t $X_{i, n-1}$, assume no doublings.

$$\text{Res } B_n = \underbrace{B_{n-1}(1, \dots, n-1)}_{\text{no } X_{i, n}} + \sum_{i=2}^{n-2} \frac{1}{X_{i, n}} C_i^{(n-5)}(1, \dots, n-1) + \sum_{2 \leq i_1 < i_2 \leq n-2} \frac{1}{X_{i_1, n} X_{i_2, n}} C_{i_1 i_2}^{(n-6)}(---)$$

$$+ \sum_{2 \leq i_1 < i_2 < i_3 \leq n-2} \frac{1}{X_{i_1, n} X_{i_2, n} X_{i_3, n}} C_{i_1 i_2 i_3}^{(n-7)}(---) + \dots +$$

$$+ \sum_{2 \leq i_1 < \dots < i_{n-4} \leq n-2} \frac{1}{X_{i_1, n} \dots X_{i_{n-4}, n}} C_{i_1 i_2 \dots i_{n-4}}^{(0)}(---)$$



not all cutsets can be used: $X_{1,n-1}$ can't be special (appears in eq with ≥ 2 terms)

→ for 1-terms this leaves us with $n-4$ cutsets

Can we pick $n-4$ of the $X_{i,n}$ as independent?

simpler decn: $(n=6)$ crossing clouds are X_{26}, X_{36}, X_{46}

$$\text{Res } B_6 = B_5(1, \dots, 5) + \sum_{i=2}^4 \frac{1}{X_{i6}} C_i^{(1)}(1, \dots, 5) + \sum_{2 \leq i_1 < i_2 \leq 4} \frac{1}{X_{i_1 6} X_{i_2 6}} C_{i_1 i_2}^{(0)}(1, \dots, 5)$$

26, 36
26, 46
36, 46

Cutsets: (+1) $X_{24} = X_{14} - X_{13}$
 $X_{25} = -X_{13}$ → good for
 $X_{26} = -X_{13}$ 36, 46

(+1) $X_{35} = X_{25} - X_{24}$ → good for
 $X_{36} = X_{26} - X_{24}$ 36, 46
 $X_{31} = -X_{24}$ 26, 46

(+2) $X_{46} = X_{36} - X_{35}$ → good for
 $X_{41} = X_{31} - X_{35}$ 26, 46
 $X_{42} = -X_{35}$ 26, 36

(+3) $0 = X_{41} - X_{46}$ → good for 26, 36
 $X_{51} = X_{41} - X_{46}$
 $X_{53} = -X_{46}$

(+4) $X_{62} = X_{52} - X_{51}$
 $X_{63} = X_{53} - X_{51}$
 $X_{64} = -X_{51}$

(+5) $X_{13} = X_{63} - X_{62}$
 $X_{14} = X_{64} - X_{62}$
 $X_{15} = -X_{62}$

No because
 X_{15} is
 special

any cutset we can select all the $C_{i,j}^{(0)}$ to zero.

next: low residue w.r.t. X_{i6}

thing that can be a problem:

assume that $n=9$. Suppose we have proved that any surviving diagram has an eer (cloud is X_{i8} for simplicity). rename eer:

$\text{Res } B_9 = \text{small diagrams that had the eer}$
 X_{i8} since there are no crossings, this looks like B_8

goal: we attempt to show that B_8 also has an eer.

potential problem:

on B₈ we cannot use all 1-zeros!
because they not derived from the B₈ constants where X₁₈
is not special! (n-5 = 4 of them, but n=8 here 5 1-zeros)

let's check:

(+4)

$$X_{62} = X_{52} - X_{51}$$

$$X_{63} = X_{53} - X_{51}$$

$$X_{64} = -X_{51}$$

(+5)

$$X_{13} = X_{63} - X_{62}$$

$$X_{14} = X_{64} - X_{62}$$

$$X_{15} = -X_{62}$$

set X₁₅ to zero:

(+4)

$$X_{62} = X_{52}$$

$$X_{63} = X_{53}$$

$$X_{64} = 0$$

→ no constant on
n=5

(+5)

$$X_{13} = X_{63}$$

$$X_{14} = X_{64}$$

$$X_{62} = 0$$

→ no constant
n=5

descent to n=5 means: get rid of eqs. that have X₁₅, X₂₆, X₃₆, X₄₆

(+0)

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = -X_{13}$$

$$-X_{26} = -X_{13}$$

descent

(+0)

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = -X_{13}$$

(+1)

$$X_{35} = X_{25} - X_{24}$$

$$-X_{36} = X_{26} - X_{24}$$

$$X_{31} = -X_{24}$$

→

(+1)

$$X_{35} = X_{25} - X_{24}$$

$$X_{31} = -X_{24}$$

(+2)

$$-X_{46} = X_{36} - X_{35}$$

$$X_{41} = X_{31} - X_{35}$$

$$X_{42} = -X_{35}$$

→

(+2)

$$X_{41} = X_{31} - X_{35}$$

$$X_{42} = -X_{35}$$

(+3)

$$0 = X_{41} - X_{46}$$

$$X_{52} = X_{42} - X_{41}$$

$$X_{53} = -X_{41}$$

→

(+3)

$$X_{52} = X_{42} - X_{41}$$

$$X_{53} = -X_{41}$$

possible starts on n=5

13 24 35 41 52

if next start is 13

move +0 and +1

↓

2 constants

if next start is 24

move +1, +2

if next start is 35

move +2, +3

(41)

only move (+3)

→ 3 left

(52)

only move (+0)

try shift introduce an X_{1n} variable and set the constant to be

$$X_{2j} = X_{1j} - X_{13} \quad j = 4, \dots, n$$

$B = 0$ when we have constant and set $X_{1i} \geq 0$
 What if we first set things to ∞ to learn only 1 item and then set $X_{1n} \geq 0$?

do these constants descend? try $n=6$

$$\begin{aligned} X_{24} &= X_{14} - X_{13} \\ X_{25} &= 0 - X_{13} \\ X_{26} &= X_{16} - X_{13} \end{aligned} \xrightarrow{n=5} \begin{aligned} X_{24} &= X_{14} - X_{13} \\ X_{25} &= -X_{13} \end{aligned}$$

maybe

debug: look at 2-zero for $n=6$ available with $X_{15}=0$, project this to $n=5$

$$\begin{aligned} X_{35} &= X_{15} - X_{14} \\ (+2) \quad X_{35} &= X_{25} - X_{24} \\ X_{36} &= -X_{14} \\ X_{36} &= X_{26} - X_{24} \end{aligned} \xrightarrow{\substack{X_{15}=0 \\ n=5}} \begin{cases} X_{35} = -X_{14} \\ X_{35} = X_{25} - X_{24} \end{cases} \leftrightarrow \begin{cases} X_{51} = X_{42} - X_{41} \\ X_{53} = -X_{41} \end{cases} \quad \begin{matrix} (+3) \\ (-2 \text{ zero}) \end{matrix}$$

$$\begin{aligned} X_{46} &= X_{26} - X_{25} \\ (+2) \quad X_{46} &= X_{36} - X_{35} \\ X_{41} &= -X_{25} \\ X_{41} &= X_{31} - X_{35} \end{aligned} \xrightarrow{\substack{X_{15}=0 \\ n=5}} \begin{cases} X_{41} = -X_{25} \\ X_{41} = X_{31} - X_{35} \end{cases} \leftrightarrow \begin{aligned} X_{41} &= -X_{25} \\ -X_{25} &= X_{31} - X_{35} \end{aligned} \quad \begin{matrix} +4 \text{ 1-zero!} \\ \text{great!} \end{matrix}$$

$n=7$

$$X_{2,4} = -X_{1,3} + X_{1,4}$$

$$X_{2,5} = -X_{1,3} + X_{1,5}$$

$$X_{2,6} = -X_{1,3} + X_{1,6}$$

$$X_{2,7} = -X_{1,3}$$

$$X_{3,5} = -X_{2,4} + X_{2,5}$$

$$X_{3,6} = -X_{2,4} + X_{2,6}$$

$$X_{3,7} = -X_{2,4} + X_{2,7}$$

$$X_{3,1} = -X_{2,4}$$

$$X_{4,6} = -X_{3,5} + X_{3,6}$$

$$X_{4,7} = -X_{3,5} + X_{3,7}$$

$$X_{4,1} = X_{3,1} - X_{3,5}$$

$$X_{4,2} = -X_{3,5}$$

$\rightarrow +0$ for $n=6$

$\rightarrow +1$ for $n=6$

$\rightarrow +2$ for $n=6$

$$X_{5,7} = -X_{4,6} + X_{4,7}$$

$$X_{5,1} = X_{4,1} - X_{4,6}$$

$$X_{5,2} = X_{4,2} - X_{4,6}$$

$$X_{5,3} = -X_{4,6}$$

$$X_{6,1} = X_{5,1} - X_{5,7}$$

$$X_{6,2} = X_{5,2} - X_{5,7}$$

$$X_{6,3} = X_{5,3} - X_{5,7}$$

$$X_{6,4} = -X_{5,7}$$

$$X_{7,2} = -X_{6,1} + X_{6,2}$$

$$X_{7,3} = -X_{6,1} + X_{6,3}$$

$$X_{7,4} = -X_{6,1} + X_{6,4}$$

$$X_{7,5} = -X_{6,1}$$

$$X_{1,3} = -X_{7,2} + X_{7,3}$$

$$X_{1,4} = -X_{7,2} + X_{7,4}$$

$$X_{1,5} = -X_{7,2} + X_{7,5}$$

$$X_{1,6} = -X_{7,2}$$

+3 for $n=6$

+4 for $n=6$

missing +5 for $n=6$

look at 2-zeros

$$X_{35} = X_{15} - X_{14}$$

$$X_{35} = X_{25} - X_{24}$$

$$X_{36} = X_{16} - X_{14}$$

$$X_{36} = X_{26} - X_{24}$$

$$X_{37} = -X_{14}$$

$$X_{37} = X_{27} - X_{24}$$

$$n=6$$

$$X_{16}=0$$



$$X_{35} = X_{15} - X_{14}$$

$$X_{35} = X_{25} - X_{24}$$

$$X_{36} = -X_{14}$$

$$X_{36} = X_{26} - X_{24}$$

$$\text{if } 25=0$$

$$X_{35} = -X_{24}$$

$$X_{35} = X_{15} - X_{14}$$

$$X_{36} = -X_{14}$$

$$X_{36} = X_{26} - X_{24}$$

look at 3-zeros

$$X_{46} = X_{16} - X_{15}$$

$$X_{46} = X_{26} - X_{25}$$

$$X_{46} = X_{36} - X_{35}$$

$$X_{47} = -X_{15}$$

$$X_{47} = X_{27} - X_{25}$$

$$X_{47} = X_{37} - X_{35}$$

$$n=6$$

$$X_{16}=0$$



$$X_{46} = -X_{15}$$

$$X_{46} = X_{26} - X_{25}$$

$$X_{46} = X_{36} - X_{35}$$

great!

new rule:

3-zeros $\equiv$ 1-zero

for $n=6$

$$\begin{aligned} (+0) \quad X_{24} &= X_{14} - X_{13} \\ X_{25} &= -X_{13} \quad \rightarrow \text{good for} \\ X_{26} &= -X_{13} \quad 36, 46 \end{aligned}$$

$$\begin{aligned} (+1) \quad X_{35} &= X_{25} - X_{24} \\ X_{36} &= X_{26} - X_{24} \\ X_{37} &= -X_{24} \end{aligned}$$

$$\begin{aligned} (+2) \quad X_{46} &= X_{36} - X_{35} \quad \rightarrow \text{good for} \\ X_{47} &= X_{37} - X_{35} \quad 26, 46 \\ X_{48} &= -X_{35} \quad 26, 36 \end{aligned}$$

$$\begin{aligned} (+3) \quad 0 &= X_{41} - X_{46} \\ X_{51} &= X_{42} - X_{46} \\ X_{53} &= -X_{46} \end{aligned}$$

$$\begin{aligned} (+4) \quad X_{62} &= X_{52} - X_{51} \\ X_{63} &= X_{53} - X_{51} \\ X_{64} &= -X_{51} \end{aligned}$$

$$\begin{aligned} (+5) \quad X_{13} &= X_{63} - X_{62} \\ X_{14} &= X_{64} - X_{62} \\ X_{15} &= -X_{62} \end{aligned}$$

$$X_{35} = X_{15} - X_{14}$$

$$X_{35} = -X_{24}$$

$$X_{36} = X_{26} - X_{24}$$

$$X_{41} = X_{15} - X_{35}$$

$$X_{42} = -X_{35}$$

x

$$X_{36} = -X_{14}$$

Q: can we prove requirement of sort using only highest k -zeros?

(can we prove that any edge is part of a triangle using only 2-zeros or only 3-zeros, etc)

Q: can we prove that nothing changes a sort using only highest k -zeros?

Q: *more important* can we recover all the zeros for $n-1$ from n with a cut sort?

true for $n=6$